

Weighting Scheme: Equal Risk Contribution (ERC)

MetaFVG Mean-Reversion Portfolio – 14-Instrument Diversified Universe

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1 Scheme Description

The portfolio blends 14 instruments trading a Spearman-gated MetaFVG mean-reversion strategy, deliberately assembled across different central banks and currency blocs (AUDUSD, EURUSD, GBPUSD, NZDUSD, USDCAD, USDCHF, USDJPY, US500, GER40, JP225, HK50, USDSGD, EURNOK, USDZAR). The current production blend uses equal weights ($w_i = 1/14$ for all i). This report tests whether replacing equal weighting with *Equal Risk Contribution* (ERC), also known as true risk parity, further reduces realized correlation and improves diversification.

Unlike inverse-volatility weighting, ERC uses the full covariance matrix $\Sigma = \text{Cov}(r)$ of daily returns, so it accounts for cross-instrument correlation rather than standalone volatility alone. The goal is to find a weight vector w , with $w_i > 0$ and $\sum_i w_i = 1$, such that every instrument contributes equally to total portfolio variance.

1.1 ERC Formulation

Total portfolio variance is $\sigma_p^2(w) = w^\top \Sigma w$. The marginal contribution to risk of asset i is

$$\text{MRC}_i = \frac{\partial \sigma_p^2}{\partial w_i} = (\Sigma w)_i,$$

and its risk contribution is

$$\text{RC}_i = w_i \cdot \text{MRC}_i = w_i (\Sigma w)_i, \quad \sum_i \text{RC}_i = \sigma_p^2(w).$$

ERC seeks w such that $\text{RC}_i = \sigma_p^2(w)/n$ for all i , i.e. every instrument contributes the same fraction of total variance.

1.2 Solve Algorithm

The weights were found with a cyclical multiplicative-update algorithm (no `scipy.optimize` dependency, and no BLAS matrix-matrix multiply – the Σw product is computed via elementwise broadcast-and-sum rather than `@` to avoid a known BLAS/LAPACK crash in this environment):

$$\begin{aligned}
w_i^{(0)} &= 1/n \\
\text{MRC}^{(k)} &= \Sigma w^{(k)} \quad (\text{elementwise broadcast} + \text{row sum}) \\
\text{RC}_i^{(k)} &= w_i^{(k)} \cdot \text{MRC}_i^{(k)} \\
\text{target}^{(k)} &= \frac{1}{n} \sum_i \text{RC}_i^{(k)} \\
\tilde{w}_i^{(k+1)} &= w_i^{(k)} \sqrt{\text{target}^{(k)} / \text{RC}_i^{(k)}} \\
w^{(k+1)} &= \tilde{w}^{(k+1)} / \sum_i \tilde{w}_i^{(k+1)}
\end{aligned}$$

iterated up to 2000 times, stopping when $\max_i \text{RC}_i - \min_i \text{RC}_i < 10^{-8} \cdot \sigma_p^2(w)$.

Convergence: the solve converged in **11 iterations**, well within the 2000-iteration budget. At convergence the spread of risk contributions relative to target, $(\max_i \text{RC}_i - \min_i \text{RC}_i) / \text{target}$, was 4.97×10^{-8} – effectively exact equalization of per-instrument risk contribution.

2 ERC Weights

Table 1: ERC weights, sorted descending. Ratio column is $w_i / (1/14)$.

Symbol	Weight	Weight / (1/14)
USDSGD	0.136 22	1.907
USDCAD	0.109 44	1.532
EURUSD	0.102 83	1.440
EURNOK	0.099 40	1.392
USDCHF	0.090 32	1.265
GBPUSD	0.079 49	1.113
USDJPY	0.074 03	1.036
AUDUSD	0.059 38	0.831
NZDUSD	0.054 15	0.758
USDZAR	0.051 09	0.715
GER40	0.040 96	0.573
US500	0.040 08	0.561
HK50	0.033 72	0.472
JP225	0.028 88	0.404
Sum	1.000 00	14.000

ERC concentrates weight in the lowest-variance G10 FX pairs (USDSGD, USDCAD, EURUSD, EURNOK, USDCHF) at roughly 1.1–1.9× the equal-weight allocation, and pulls weight away from the highest-variance instruments – the equity indices (JP225, HK50, US500, GER40) and the more volatile crosses (NZDUSD, USDZAR, AUDUSD) – down to roughly 0.4–0.8× equal weight.

Table 2: Realized correlation and diversification impact, equal-weight vs. ERC.

Metric	Equal Weight	ERC
Weighted avg. pairwise correlation	0.015 615	0.014 191
Diversification ratio (DR)	2.080 922	1.830 624

3 Correlation Impact

ERC lowers the weighted average pairwise correlation slightly ($0.0156 \rightarrow 0.0142$, a $\sim 9\%$ relative reduction) by tilting away from instruments that co-move more with the rest of the book. However, the diversification ratio *falls* ($2.081 \rightarrow 1.831$): DR rewards spreading weight across instruments with high *standalone* volatility relative to the blended portfolio’s volatility, and ERC does the opposite by design – it deliberately down-weights the highest-standalone-vol instruments (equity indices, JP225/HK50/US500/GER40) to equalize their risk *contribution*. The two diversification metrics move in opposite directions here.

4 Results

Table 3: Quantstats metrics, equal-weight vs. ERC blend.

Metric	Equal Weight	ERC
Sharpe	0.517 941	0.311 482
Sortino	0.816 977	0.487 811
Calmar	0.343 017	0.191 157
CAGR	0.000 183	0.000 102
Max Drawdown (%)	−5.325 228	−5.325 228
Volatility (%)	3.527 622	3.269 655
Tail Ratio	1.238 689	1.114 092
Ulcer Index	0.000 192	0.000 183
Kelly Criterion	0.066 763	0.045 890
VaR (%)	−0.358 268	−0.334 747
CVaR (%)	−0.715 011	−0.733 195
Skew	1.544 839	1.948 282
Recovery Factor	2.260 123	1.259 807
Win rate	0.308 850	0.304 636
Best day	0.000 291	0.000 291
Worst day	−0.000 239	−0.000 239
Total return	0.001 204	0.000 671

5 Takeaway

ERC modestly reduces weighted average pairwise correlation (-9% relative) but this does not translate into a better risk-adjusted portfolio: Sharpe falls from 0.518 to 0.311, Sortino from 0.817 to

0.488, and total return drops by more than 40%, while maximum drawdown is unchanged. The diversification ratio actually *worsens* ($2.081 \rightarrow 1.831$), because ERC's risk-contribution-equalizing objective directly conflicts with DR's standalone-volatility-weighted definition of diversification in this universe. On this data, equal weighting is the better blend – ERC trades away return concentrated in the equity-index legs (which it underweights) for a small, not-obviously-worthwhile correlation reduction.